LEARNER GUIDE

Recurring decimals as fractions

Number Systems and Calculator Judgement · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Every fraction gives a decimal that either **terminates** (3/8 = 0.375) or **recurs** (1/3 = 0.333…, written $0.\dot{3}$). When you divide by 7, the only possible remainders are 0 to 6, so if the division never ends a remainder must come back — and then the digits repeat. To turn a recurring decimal into a fraction, multiply it by 10, 100 or 1000 so that the repeating tails line up, then subtract: the tails cancel.

What you are learning to notice

- Simplify first. A fraction terminates when its denominator's only prime factors are 2 and 5.
- Dot notation: one dot over a single repeating digit ($0.\dot{3}$); dots over the first and last digits of a repeating block ($0.\dot{4}\dot{5}$ = 0.4545…).
- Multiply by 10 if one digit repeats, by 100 if two repeat, by 1000 if three repeat.
- $0.\dot{9}$ = 1 exactly: the method gives $9x = 9$.

Moving through it, one step at a time

Let x be the recurring decimal. Multiply by the power of 10 that shifts the decimal point by one repeating block, and write it under x so the tails line up. Subtract to cancel the tails, divide, then simplify. Check by dividing the numerator by the denominator.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Why the digits repeat

Work out 3/7 as a decimal, watching the remainders.

1. 30 ÷ 7 = 4 r 2, then 20 ÷ 7 = 2 r 6, 60 ÷ 7 = 8 r 4 …	digits 4, 2, 8 …
2. … 40 ÷ 7 = 5 r 5, 50 ÷ 7 = 7 r 1, 10 ÷ 7 = 1 r 3	digits 5, 7, 1
3. Remainder 3 is where we started, so everything repeats.	0.428571428571…

3/7 = $0.\dot{4}2857\dot{1}$. Only 6 non-zero remainders are possible, so the block has at most 6 digits.

Example 2 — Line up the tails

Write $0.\dot{4}\dot{5}$ (0.4545…) as a fraction.

1. Two digits repeat, so multiply by 100.	x = 0.4545…, $100x$ = 45.4545…
2. Subtract: the tails cancel.	$100x - x = 45$, so $99x = 45$
3. Divide and simplify.	$x = 45/99 = 5/11$
4. Check: 5 ÷ 11 = 0.4545… ✓	✓

$0.\dot{4}\dot{5}$ = 5/11.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (changing fractions to decimals by division, and simplifying fractions), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.